HUSSEIN NSANZIMFURA

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Final-year Electrical Power Engineering student at the University of Rwanda focused on applying machine learning to power system reliability and renewable energy in Africa. Built projects on electricity theft detection, solar microgrid design and transformer health monitoring, and shared a research preprint on non-technical losses. Winner of the AI for Public Good hackathon. Completed the CMU-Africa Bridge Program. Interested in graduate research on data-driven approaches to grid reliability and clean energy access in Africa.

EDUCATION

Bachelor of Science in Electrical Power Engineering | University of Rwanda, College of Science and Technology | 2022 - 2027 (expected)

- Cumulative GPA: 3.22/4.00
- Coursework: Power Electronics, Electrical Machines, Electrical Energy Utilization, Embedded Systems, Control Systems

EXPERIENCE

Jabana Thermal Power Plant, Industrial Training | Rwanda | Academic year 2025/2026

- Observed thermal power generation and plant operations, and studied the electrical equipment and safety practices used on site.

REG EUCL, Industrial Training at Mount Kigali Substation | Rwanda | Academic year 2026/2027

- Studied the control power systems that keep protection and control equipment running at the substation.
- Used access to the National Control Centre system to observe how grid operations are monitored and controlled.

ENGINEERING PROJECTS AND RESEARCH

Machine Learning-Based Electricity Theft Detection System | GitHub

- Built an XGBoost pipeline on smart meter data from 42,372 customers (1,035 days, 8.5% thieves) using 15 engineered behavioural features. Reached ROC-AUC 0.72 and about 44% theft recall, over 2x better than random inspection, and deployed it as a live Streamlit demo.
- Technologies: Python, Pandas, Scikit-learn, XGBoost, Streamlit.

Solar Microgrid Design for a Rural Health Centre in Rwanda | GitHub

- Designed a 7 kWp solar PV and 40 kWh LiFePO4 battery microgrid (6 kW inverter, 48 V DC) for a rural health centre with a 20 kWh/day load. Python simulation showed 0 kWh of unmet load; solar LCOE of $0.364/kWh against $0.727/kWh for diesel, saving about $70.6k over 25 years.
- Technologies: Python, NumPy, Matplotlib, PVGIS data.

Machine Learning-Based Transformer Health Monitoring and Predictive Maintenance | GitHub

- Ongoing project building machine learning models for fault detection and predictive maintenance of distribution transformers, using public dissolved gas analysis datasets and synthetic data reflecting Rwandan conditions.
- Technologies: Python, machine learning, data analysis.

Research Paper: Machine Learning-Based Detection of Non-Technical Losses for Improving Distribution Grid Reliability: A Case Study Using Smart Meter Data. Preprint on ResearchGate, 2026. View on ResearchGate

TRAINING AND PROFESSIONAL DEVELOPMENT

- **CMU-Africa Bridge Program** (Spring 2026): six-week program in machine learning, IoT, embedded systems, Arduino and academic communication. Completed.
- **AFRETEC Network:** Artificial Intelligence, Machine Learning and Big Data Analytics.
- **Kigali Smart Skills Hub, Rwanda Polytechnic Kigali College:** IoT and AI Model Development (ongoing).
- **ICT Chamber Rwanda:** Data Science Training.
- **Aspire Leaders Program:** Alumni.

LEADERSHIP AND COMMUNITY

Campus Co-Director, Millennium Fellowship (Class of 2026) | University of Rwanda, Nyarugenge Campus

- Lead campus activities linking student engagement to the UN Sustainable Development Goals; engaged with the Sustainable Microgrid Initiative on renewable energy access.

Former Collaboration Tools Group Leader | Empower Forum Association of Electrical Power Engineering Students, UR-CST

- Coordinated collaboration and knowledge-sharing among electrical power engineering students.

Active Member | IEEE Rwanda Subsection

AWARDS

Winner, AI for Public Good Hackathon | View

TECHNICAL SKILLS

- **Electrical Power Engineering:** Power systems, electricity distribution, renewable energy, solar PV, microgrids, transformer monitoring
- **Programming and Data:** Python, Pandas, NumPy, Scikit-learn, XGBoost, machine learning fundamentals, Power BI
- **Tools:** MATLAB/Simulink, AutoCAD Electrical, Arduino, GitHub, Google Colab, Streamlit
- **Professional:** Technical research and writing, engineering problem-solving, teamwork
- **Languages:** Kinyarwanda, English, French (working)

CONFERENCES

Nuclear Energy Innovation Summit for Africa (NEISA) | Africa Energy Export | Africa Simulation Conference | Rwanda Youth Nuclear Summit | Empower Forum events on renewable energy